

## Quiz 2

Course: CSE422

Time: 30 minutes

**Student ID:** \_\_\_\_\_

**Section:** \_\_\_\_\_

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**Question 1: Linear Regression and Gradient Descent****[10 Marks]**

Suppose you are asked to predict a person's weight based on the person's age and height. You notice that these variables have an approximately linear relationship, so you choose linear regression to solve the problem.

The training dataset is given below:

| Sl. | Age ( $x_1$ ) | Height ( $x_2$ ) | Weight ( $y_i$ ) |
|-----|---------------|------------------|------------------|
| 1   | 2             | 3                | 4                |
| 2   | 3             | 4                | 5                |
| 3   | 4             | 5                | 6                |

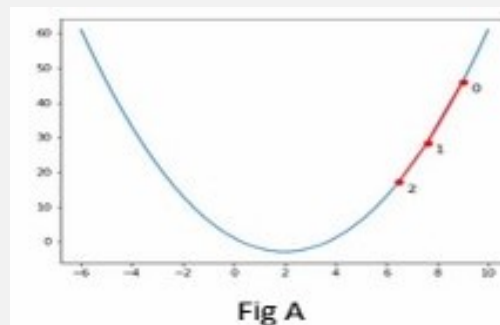
The hypothesis function is:

$$y'_i = m_1x_1 + m_2x_2 + c$$

The loss function for the regression model is:

$$L = \frac{1}{N} \sum_{i=1}^N (y_i - y'_i)^2$$

- Describe the purpose of the learning rate. [1]
- In your opinion, does gradient descent always guarantee finding the global minimum? Briefly justify your answer. [2]



**Fig A: Loss vs Coefficient**

- Suppose Figure A represents a loss function curve, where the  $y$ -axis represents the loss and the  $x$ -axis represents the coefficient value. The curve shows how gradient descent slowly optimizes the coefficient and moves it toward the point of global minimum.

At some iteration  $n$ , assume the model is at point 1. In the next iteration, gradient descent moves the model to point 2 and not point 0. Explain how gradient descent performs this update. [3]

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- Assume learning rate  $\alpha = 0.1$ ,  $m_1 = 0.5$ ,  $m_2 = 0.4$ , and  $c = 0.6$ . Using a single iteration of gradient descent, estimate the new values of  $m_1$ ,  $m_2$ , and  $c$ . [4]